



Assessment of different configurations of solar energy driven organic flash cycles (OFCs) via exergy and exergoeconomic methodologies



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ABSTRACT

Different configurations of organic flash cycles (OFCs), including simple OFC, dual flash OFC, improved OFC, expander-OFC and improved expander-OFC, analyzed from the viewpoints of exergy and exergoeconomic. Five appropriate working fluid including Toluene, p-Xylene, m-Xylene, Ethylbenzene and o-Xylene are considered as working fluid. The effect of flashing temperature is investigated on the exergy and exergoeconomic performance of the system. Also, an optimization procedure is carried out to find the most efficient configuration and optimum flashing temperature from the viewpoints of exergy and exergoeconomic. Results show that the minimum and maximum optimal exergy efficiency belongs to the simple OFC and improved expander-OFC, respectively. Also, the lowest and the highest minimum unit cost of produced power refers to the simple OFC and dual flash OFC. Moreover, the results indicate that Toluene as working fluid has the best performance from the viewpoints of both exergy and exergoeconomic. Based on the results, expander-OFC is chosen as the configuration that satisfies both exergy and economic criteria, simultaneously. Finally, an optimal design point for expander-OFC is obtained from Pareto frontier with the exergy efficiency of 49.1% and unit cost of produced power of 31.4 \$/GJ which occurs at the flashing temperature of 427.6 K.

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1. Introduction

Fossil fuels limitation in a world with increasing energy consumption and population encourage researchers to provide novel systems of renewable energy using [1,2]. From the other hand, introducing efficient energy converting systems powered by low and medium temperature heat sources plays an important role in the energy policy of developed societies [3]. Among the all low and medium temperature heat source powered cycles, organic Rankine cycles (ORCs) have some manifest advantages [4]. In the recent years, use of medium temperature heat sources to produce power, ORC has attracted the attention of scientists because of its simplicity, flexibility and reliability [5,6]. As a fact, organic fluids (which are employed in ORCs as working fluid) have less enthalpy drop through the expander, in comparison to the water in Rankine cycle and this is necessitating the use of higher mass flow rates. A higher mass flow rate through the ORC expander declines the gap losses, which in turn increases the turbine adiabatic efficiency [7,8].

Besides the mentioned advantage of ORCs, the main disadvantage of these cycles is high exergy destruction at the evaporator. Various strategies are presented to cover the mentioned problem, i.e. using zeotropic mixtures working fluids [9,10], employment of transcritical cycles [11,12], organic flash cycles (OFCs) [13,14]. OFCs are promising technology, which attracts researchers' attention. Mainly, good temperature matching and reducing exergy destruction during heat addition is the aim of OFCs introducing. A large number of research works have been reported about OFCs and ORCs with the aim of heat recovering. Following a literature review has included representing the authors' finding in this field.

Ho et al. presented a comparative study between OFC and other advanced vapor cycles including, a zeotropic Rankine cycle using a binary NH₃/H₂O mixture, an optimized basic organic Rankine cycle (ORC), and a transcritical CO₂ cycle. They used 10 different aromatic hydrocarbons and siloxanes as potential working fluids. They concluded that aromatic hydrocarbons have the best performance as the working fluid for the ORC and OFC due to much produced power and less complexity of turbine designs. Also, they showed that the single flash OFC reaches comparable utilization efficiencies to the optimized basic ORC [14]. The same authors in another investigation presented different configurations of OFCs with the

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